HEALTH360

Clinical & Healthcare Risk Analysis

Final Project Report

Project Date: 19 October 2026

Dataset: 5,000 records | 17 columns | Tableau-based analysis

Project Website

Health360 Web Application

Health360 Web Application

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1. INTRODUCTION

1.1 Project Overview

Health360: Clinical & Healthcare Risk Analysis is a data-visualization project designed to make a healthcare dataset easier to explore. The project focuses on descriptive analysis of patient demographics, clinical indicators, lifestyle behaviours and recorded disease outcomes. The project brief specifies 5,000 patient records across 17 columns. Tableau visualizations are presented through a web page that provides access to the dashboard and story.

The project examines group-level patterns involving age group, average blood pressure, BMI category, heart disease, diabetes, exercise level, fatigue, alcohol consumption and smoking. It is intended to support exploration and communication of data, rather than to diagnose an individual or replace professional medical judgement.

1.2 Purpose

- Present healthcare data through clear and accessible visualizations.
- Compare patient groups by age and BMI category.
- Explore relationships between lifestyle indicators and recorded health outcomes.
- Provide a dashboard and story that help communicate findings.
- Support data-informed discussion of preventive-care priorities.

2. IDEATION PHASE

2.1 Problem Statement

Healthcare datasets can contain multiple demographic, clinical and lifestyle variables. Reviewing thousands of records manually can make it difficult to compare patient groups and identify descriptive patterns quickly. Health360 addresses this analysis and communication challenge by organizing selected measures into Tableau charts and presenting them through a website.

2.2 Empathy Map Canvas

Target users include healthcare providers, clinical risk assessment officers, healthcare data analysts and public-health researchers.

- Thinks: Which patient groups show different patterns in blood pressure, BMI or recorded disease outcomes?
- Needs: A clear way to compare demographic, clinical and lifestyle measures.
- Sees: A dataset with 5,000 records and multiple variables that can be difficult to review manually.
- Does: Uses charts, comparisons and a Tableau story to explore and communicate observations.
- Pain points: Time-consuming manual review and difficulty comparing multiple variables.
- Desired gain: Faster visual exploration and a clearer basis for group-level discussion.

2.3 Brainstorming

- Create a KPI overview for a quick view of patient and disease-related metrics.
- Compare average blood pressure across six age groups.
- Compare alcohol and smoking measures across age groups.
- Show exercise-level and age-group distributions using donut and pie charts.
- Compare heart disease and diabetes across BMI categories.
- Use a heat map to explore heart-disease patterns by BMI and exercise level.
- Show fatigue-condition distribution across age groups.
- Publish the dashboard and story through the Health360 website.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

3.2 Solution Requirement

- Display project overview and purpose.
- Provide access to the interactive Tableau dashboard and Tableau story.
- Present the nine chart types identified in the project brief.
- Support readable navigation across website sections.
- Work in a modern web browser with internet access for embedded Tableau content.
- Make it clear that the visualizations show descriptive patterns, not validated clinical predictions.

3.3 Data Flow Diagram

High-level data flow:

Healthcare Dataset (5,000 records, 17 columns) → Tableau Data Connection → Data Analysis & Visualizations → Tableau Dashboard / Story → Health360 Website → User

The dataset is imported into Tableau, analyzed through charts and then exposed through the website using Tableau's public embed links. This diagram is a high-level representation based on the project description; the screenshots and links do not specify additional server-side processing.

3.4 Technology Stack

4. PROJECT DESIGN

4.1 Problem-Solution Fit

Customer segments: healthcare providers, clinical risk assessment officers, healthcare data analysts and public-health researchers. Core problem: manual inspection of multiple patient indicators makes group comparisons difficult. Constraints: time required for manual analysis and the need to interpret many variables together. Proposed fit: an accessible website with Tableau visualizations that helps users compare age groups, BMI categories, lifestyle measures and recorded disease outcomes.

4.2 Proposed Solution

Health360 provides a website that links users to a Tableau dashboard and story. The supplied project brief lists nine visualization types:

– KPI — Patient Health & Disease Overview

– Bar chart — Average Blood Pressure Across Age Groups

– Grouped bar chart — Alcohol and Smoking Consumption Across Age Groups

– Donut chart — Exercise Level Distribution

– Pie chart — Patient Distribution by Age Group

– Grouped bar chart — Heart Disease vs Diabetes Prevalence by BMI Category

– Bar chart — BMI Category vs Heart Disease Rate

– Heat map — Heart Disease by BMI Category and Exercise Level

– Bar chart — Age Group Distribution by Fatigue Condition

4.3 Solution Architecture

• Data layer: healthcare dataset described as 5,000 records and 17 columns.

• Analytics layer: Tableau connects to the data and provides the charts, dashboard and story.

• Presentation layer: Health360 web page embeds or links to the Tableau Public content.

• Access layer: users open the website in a browser and navigate to the visualizations, dashboard or story.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

A dated task-by-task schedule was not provided in the source material; the phases above represent a logical project workflow rather than a claim about actual completion dates.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

No measured load-time, throughput, response-time or stress-test results were supplied. Therefore, this report does not claim numerical performance results.

7. RESULTS

The project brief and supplied Tableau screenshots document a healthcare analysis dashboard with views for average blood pressure by age group, lifestyle measures, age-group distribution, exercise-level distribution, BMI-related disease comparisons and fatigue condition. The supplied project description reports that the dataset contains 5,000 records across 17 columns.

7.1 Output Screenshots

Insert current screenshots of the Health360 website Overview, Visualizations page, Tableau dashboard and Tableau story here before final submission. Screenshots were not embedded in this report because the supplied images show the reference/template rather than clean exported application screenshots.

- Health360 website
- Tableau dashboard
- Tableau story

8. ADVANTAGES & DISADVANTAGES

Advantages

- Visual summaries can make large datasets easier to explore.
- Multiple charts support comparisons across age groups and BMI categories.
- Dashboard and story provide a structured way to present findings.
- Website and GitHub Pages provide a shareable access point.

Disadvantages / Limitations

- The analysis depends on the completeness and quality of the source dataset.
- Visual associations do not establish cause and effect.
- Tableau Public embeds require internet access and may be affected by browser/embed restrictions.
- The described charts should not be used as a validated medical diagnostic or risk-prediction system.
- No quantitative performance-test results were provided.

9. CONCLUSION

Health360 demonstrates how Tableau visualizations can be presented through a website to explore healthcare data. Using the stated dataset of 5,000 records and 17 columns, the project organizes nine visualization types covering patient overview, blood pressure, lifestyle indicators, age distribution, BMI-related disease patterns and fatigue. The dashboard and story support group-level exploration and communication. The results are descriptive and should be interpreted alongside data-quality considerations and professional clinical judgement.

10. FUTURE SCOPE

- Add interactive filters for age, BMI, gender and lifestyle variables where supported by the dataset.
- Improve accessibility, mobile layout and embed-loading feedback.
- Add documented data definitions and source/provenance notes.
- Include validated statistical analysis and confidence intervals where appropriate.

- If a predictive model is introduced, validate it on suitable data and assess bias, calibration and clinical safety before any real-world use.
- Add automated data refresh and monitoring if the dataset and hosting setup support it.

11. APPENDIX

Source Code / GitHub Repository

https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git

Dataset Link

A separate public dataset URL was not provided. The project brief describes a smart healthcare dataset with 5,000 records and 17 columns. Add the exact dataset source or repository file link here if it is shareable.

Project Demo / Web Page

https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/

Tableau Dashboard

Open Tableau dashboard

Tableau Story

Open Tableau story

Note: The dashboard and story links supplied for this report are identical. If they are intended to open different Tableau views, replace the story link with the correct story-specific URL before submission.

Project Links

GitHub repository: https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git

Website: https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/

Tableau dashboard: Open Tableau dashboard

Tableau story: Open Tableau story

Note: The dashboard and story URLs supplied were identical. Replace the story URL if a separate story-specific link is available.